

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EMT 326 — Past Questions & Materials
by Renance

Course: EMT 326
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 300

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Paper 1 of 2 — EMT 326 Wildlife Ecology & Management 2024

Level 300 • First Semester • PastQuestion

ao: FEDERAL UNIVERSITY OF PETROLEUM RESOURCES

: COLLEGE OF SCIENCE

DEPARTMENT OF ENVIRONMENTAL MANAGEMENT AND TOXICOLOGY

FIRST SEMESTER 2022/2023 EXAMINATION

EMT326: WILDLIFE ECOLOGY AND MANAGEMENT (3 units) Time: 3 hours

INSTRUCTION: Answer four (43 questions out of the six (6) provided.

egradation due to human activities.

tem services. (b) Explain the role of

ment, highlighting their

4. (a) A national park is experiencing habitat d

Explain the impact on biodiversity and ecosys

plants and animals as wildlife components of the environ

conibui'ons to ecosystem balance.

2. (a) Explain the concept of energy flow in ecosystems, using the 10% rule to

'ilustrate the inefficiency of energy transfer. (b) Describe the components of an

ecosystem, highlighting the interactions between biotic and abiotic factors.

3. (a) Explain the characteristics of wildlife population. (b) Explain the factors that

influence wildlife movement x

4. (a) How does anima! behaviour correlate with its adaptation ability. (b) Briefly

discuss the term wildlife ecology or wildlife habitat. : :

5. (a) List drug classes used in wild life and explain any two. (b) Give five reasons

for wiid life capture. ae

explain any four. (b) As an

6. (a) List six wild life capturing techniques and

ndling captured animals from

environmentalist what are the procedure of ha

stress.

Good luck !.

Paper 2 of 2 — EMT 326

Level 300 • Second Semester • Handout

WILDLIFE POPULATION

All non domesticated plants, animals and other species are considered as wildlife.

Population is the total number of specific species in a particular area. All forms of wildlife experience swing in animal populations in response to changes in habitat, hunting trapping and weather.

The changes in the population size and demographic over time is known as population dynamics

Characteristics of wildlife population

1. Population size: this is the total number of people in a population.

Some factors affecting population size are

- a. Birth rate: this refers to the number of birth in a given time period
- b. Death rate: this is the number of death in a given time period
- c. Immigration: this refers to the number of people moving into the population
- d. Emigration: : this refers to the number of people moving out of the population

2. Population density: this is the number of people in a region or per unit area. The amount of competition increases with population density. It can be crude (total number of people per unit area or ecological(total number of people per unit of a suitable habitat). Population density help to show how crowded a population is and can affect birth and death rate

3. Population dispersion/distribution: this is the geographic or clustering of the member of a population. The distribution pattern can be random (in which individual are spread randomly with in an area) or uniform (in which individual are evenly spaced due to territorial behavior). Some factors affecting dispersion are, availability of resources, social interaction and environmental condition.

4. Sex distribution: this is the percentage of male and female in a population. The highest possible birth rate can be calculated using population dynamics by calculating the sex ratio (proportion of male to female in the population) the sex ratio affect matting pattern and future population growth

5. Age distribution: this is the number of individual in a population who fall within a given age range and who are organized into a class or cohort (cohort is a group of people in a population who were born around the same time). It shows the reproductive status and potential for future population growth. The age categories are

Pre- reproductive age

Reproductive age

Post - reproductive age

WILD LIFE MORTALITY

Wild life mortality refers to the death of wild life from various causes.

Understanding wildlife mortality is essential for conservation as it helps to identify threat to species and ecosystem.

Types of mortality

Wild life mortality can be grouped into two types based

On their cause into

a. Natural mortality: these are death caused by natural factors such as, predation: This is death caused by being preyed upon by other animals

Disease: this is mortality caused by infectious agents such as (viruses, bacteria, parasites) or non infectious diseases.

Starvation: this death is due to lack of sufficient food resources

Senescence: this is mortality due to old age and decline in physiological function.

b. Anthropogenic mortality: this is death resulting from human activities including habitat destruction, pollution, hunting and collision with vehicle.

Habitat destruction: this is loss of life due to habitat fragmentation, deforestation, urbanization and agriculture

Pollution: this is death caused by polluted like pesticide, heavy metal, plastic and other toxic substances.

Hunting and poaching: this mortality from legal and illegal hunting activities.

Wildlife mortality can also be cause by environmental factors like flood, wildfire, and drought.

Measuring wildlife mortality

Different method can be used to measure wildlife. Some methods are

- a. Monitoring through field observation, survey, tracking and tagging.
- b. Use of camera traps and remote sensing technologies.
- c. Analyzing carcass counts and cause of death
- d. Using population model to estimate mortality rate

EMT326:

WILDLIFE ECOLOGY AND MANAGEMENT

Introduction to Ecosystems

A community of living and non-living components interacting in a specific environment defines an ecosystem. This delicate balance of interactions is exemplified in a forest ecosystem, where living components like trees, animals, and microorganisms coexist with non-living components like soil, water, and sunlight. The trees provide shelter and food for the animals, while the microorganisms decompose organic matter, recycling nutrients. The soil supports plant growth, and the water cycle nourishes the entire ecosystem. This intricate web of relationships sustains life and maintains the forest's biodiversity.

Similarly, a coral reef ecosystem thrives in a specific marine environment, where living components like coral, fish, and algae interact with non-living components like water, salinity, and temperature. The coral provides a habitat for the fish and other marine life, while the algae photosynthesize, producing nutrients for the ecosystem. The water's chemical composition and temperature support the coral's growth, and the reef's structure protects the marine life from predators and currents. This harmonious balance of interactions makes coral reefs some of the most diverse ecosystems on the planet, supporting a vast array of marine life. Understanding these complex relationships is crucial for managing and conserving ecosystems like forests and coral reefs.

Components of an Ecosystem

The components of an ecosystem can be divided into two main categories: biotic and abiotic components.

Biotic Components:

1. Producers (Autotrophs): Plants, algae, and some bacteria that convert sunlight into energy through photosynthesis.
2. Consumers (Heterotrophs): Animals, fungi, and some bacteria that obtain energy by consuming other organisms or organic matter.
 - a. Primary Consumers (Herbivores): Feed on producers.
 - b. Secondary Consumers (Carnivores): Feed on primary consumers.
 - c. Tertiary Consumers (Top Carnivores): Feed on secondary consumers.
 - d. Decomposers: Break down dead organic matter into simple nutrients.
3. Microorganisms: Bacteria, fungi, and protozoa that play a crucial role in decomposition, nutrient cycling, and disease dynamics.

Abiotic Components:

1. Light: Solar radiation that supports photosynthesis and influences ecosystem processes.

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2. Temperature: Affects metabolic rates, growth, and distribution of organisms.
3. Water: Essential for life, influences ecosystem processes, and affects nutrient availability.
4. Nutrients: Essential elements like carbon, nitrogen, phosphorus, and potassium that support life.
5. Soil: Provides support, water, and nutrients for plant growth.

6. Topography: Influences ecosystem processes, such as water flow and soil erosion.
7. Atmosphere: Comprises gases like oxygen, carbon dioxide, and nitrogen that support life.

Types of Ecosystems

These are 3 types of ecosystems;

1. Terrestrial ecosystems (forests, grasslands, deserts)
2. Freshwater ecosystems (rivers, lakes, wetlands)
3. Marine ecosystems (coral reefs, estuaries, open ocean)
4. Artificial Ecosystems (urban ecosystems, aquaculture, zoological parks)
5. Other Ecosystems (caves, islands, alpine)

Write brief notes on each in your notes

Energy Flow in Ecosystems

These are the levels of energy flow;

- ☐ Producers (plants, algae) convert sunlight into energy
- ☐ Consumers (herbivores, carnivores) feed on producers or other consumers
- ☐ Decomposers (microorganisms) break down dead organic matter

Energy flow in ecosystems refers to the transfer of energy from one trophic level to another. It begins with producers (plants, algae, and some bacteria) that convert sunlight into energy through photosynthesis. This energy is then transferred to primary consumers (herbivores) that feed on producers, and subsequently to secondary consumers (carnivores) that feed on primary consumers. Decomposers break down dead organic matter, releasing nutrients back into the ecosystem. Energy flow is unidirectional, from sunlight to producers to consumers, with energy lost at each trophic level.

The energy flow process can be illustrated using the 10% rule, where only 10% of energy is transferred from one trophic level to the next. For example, if producers capture 100 units of energy, primary consumers receive 10 units, and secondary consumers receive 1 unit. This inefficiency explains why ecosystems have fewer top predators than producers. Understanding energy flow is crucial for managing ecosystems, as it reveals the interconnectedness of species and the importance of maintaining a balanced food

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web. Energy flow also highlights the critical role of producers in supporting life in ecosystems.

Nutrient Cycling in Ecosystems

Nutrient cycling in ecosystems refers to the continuous process of nutrient exchange between living organisms (biotic factors) and the environment (abiotic factors). This process involves the uptake, utilization, release, and transformation of nutrients. Plants absorb nutrients from the soil, water, or air, which are then used for growth, maintenance, and reproduction.

Nutrient cycling in ecosystems involves four key processes: photosynthesis,

decomposition, nutrient uptake, and nutrient release. Photosynthesis converts sunlight, CO₂, and H₂O into glucose and O₂, storing energy in organic compounds.

Decomposition, carried out by microorganisms, breaks down dead organic matter, releasing nutrients back into the environment. Nutrient uptake occurs when plants absorb essential nutrients from soil, water, or air, utilizing them for growth and maintenance.

Finally, nutrient release returns nutrients to the environment through excretion, decomposition, or death, making them available to other organisms, thereby maintaining the delicate balance of essential nutrients necessary to sustain life in ecosystems.

As organisms release nutrients through excretion, decomposition, or death, microorganisms convert them into different forms, making them available to other organisms. This cycling of nutrients maintains ecosystem balance, supports plant growth and productivity, regulates water quality, influences climate change, and affects human health and well-being.

Factors such as climate, soil type and quality, vegetation, microbial activity, and human activities (agriculture, pollution, land use changes) can impact nutrient cycling.

Understanding these processes is crucial for managing ecosystems, mitigating environmental impacts, and ensuring sustainable resource use. By recognizing the importance of nutrient cycling, we can work towards maintaining healthy and productive ecosystems.

Ecosystem Services

1. Provisioning Services

Provisioning services provide essential resources for human survival and well-being. These services include food, water, timber, and minerals, which are obtained from ecosystems through agriculture, forestry, fishing, and mining. Ecosystems like forests, grasslands, and coral reefs provide a range of provisioning services, supporting human livelihoods and economic development.

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2. Regulating Services

Regulating services maintain the health and resilience of ecosystems, regulating essential processes like climate, water, and soil quality. These services include climate regulation, pollination, water filtration, and pest control, which are crucial for maintaining ecosystem balance and human well-being. Regulating services also mitigate natural disasters like floods, landslides, and storms, protecting human settlements and infrastructure.

3. Cultural Services

Cultural services provide non-material benefits that enrich human lives, including recreation, tourism, spiritual values, and cultural heritage. Ecosystems like national parks, forests, and wetlands offer opportunities for recreation, tourism, and spiritual enrichment, supporting human well-being and cultural diversity. Cultural services

also inspire art, literature, and education, fostering creativity and innovation.

Human Impact on Ecosystems

The following are how humans impact on the ecosystem;

1. Deforestation: Clearance of forests for agriculture, urbanization, and logging, leading to habitat loss and biodiversity decline.
2. Climate Change: Greenhouse gas emissions alter global temperatures, disrupting ecosystems and causing rising sea levels.
3. Pollution: Release of toxic chemicals, plastics, and waste into air, water, and soil, harming organisms and ecosystems.
4. Overfishing: Excessive fishing depletes fish populations, damaging marine food chains and ecosystems.
5. Habitat Fragmentation: Human activities split habitats, isolating species and reducing genetic diversity.
6. Invasive Species: Introduction of non-native species outcompetes native species for resources, altering ecosystem balance.
7. Soil Degradation: Soil erosion, salinization, and nutrient depletion reduce fertility and ecosystem productivity.
8. Water Pollution: Contamination of freshwater sources harms aquatic life and human health.

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9. Overconsumption of Resources: Unsustainable use of resources like water, energy, and minerals strains ecosystems.
10. Land Conversion: Conversion of natural habitats to agricultural land, urban areas, and infrastructure reduces ecosystem diversity and function.

Plants and Animals as Wildlife Components of the Environments.

Plants and animals are vital components of the environment, playing crucial roles in maintaining ecosystem balance and diversity. Plants, as producers, convert sunlight into energy through photosynthesis, provide habitat and food for animals, release oxygen, stabilize soil, and regulate water cycles. Animals, as consumers, feed on plants or other animals, decompose organic matter, pollinate plants, disperse seeds, and modify environments to create habitats for other species. Together, plants and animals form intricate relationships, shaping ecosystems and supporting life on Earth, with plants providing the foundation for food webs and animals facilitating nutrient cycling, seed dispersal, and ecosystem engineering.

Factors Affecting Distribution and Abundance of Wild Life.

1. Habitat: The availability, quality, and connectivity of habitat determines the distribution and abundance of wildlife, as it provides essential resources such as food, water, shelter, and breeding grounds.

2. Climate: Changes in temperature and precipitation patterns, as well as extreme weather events, impact wildlife distribution and abundance by altering the availability of resources and habitats.
3. Geography: Topography, altitude, and landforms influence wildlife movement and habitat use by creating barriers, corridors, and unique microclimates that affect species distribution.
4. Food availability: The presence, abundance, and quality of food sources affect wildlife population sizes, distribution, and behavior, as species adapt to exploit available resources.
5. Predation: Predator-prey dynamics regulate wildlife populations and influence distribution by controlling prey numbers, altering behavior, and shaping evolutionary adaptations.
6. Human activities: Habitat destruction, fragmentation, degradation, hunting, poaching, pollution, and climate change reduce wildlife populations, alter distribution, and disrupt ecosystem processes.

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7. Disease and parasites: Outbreaks of disease and parasites impact wildlife population sizes, distribution, and behavior by reducing individual fitness, altering population dynamics, and influencing species interactions.
8. Evolutionary history: Adaptation to changing environments, species interactions, and evolutionary innovations influence wildlife distribution and abundance by shaping species traits, ecological niches, and community composition.
9. Species interactions: Interactions with other species, including competition, symbiosis, mutualism, and commensalism, affect wildlife distribution, abundance, behavior, and ecosystem processes.
10. Environmental changes: Natural disasters, climate change, invasive species, and other disturbances alter wildlife distribution and abundance by modifying habitats, disrupting ecosystem processes, and creating opportunities for range shifts and community reorganization.

Good luck, guys !

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